

Bhumi Dastanwala

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Education

Sardar Vallabhbhai National Institute of Technology, Surat B.Tech in Computer Science	Aug 2024 – May 2028 CGPA: 8.25
Bhulka Vihar School, Surat Higher Secondary Education	Jun 2022 – Jun 2024 90.15%
C. C. Shah School, Surat Secondary Education	Jun 2021 – Jun 2022 94.66%

Projects

AI-Powered Sudoku Intelligence Platform *Python, Genetic Algorithms, OpenCV, OCR, NumPy, Tkinter*

- Developed an AI-based Sudoku solving platform using a **Genetic Algorithm** with fitness-driven optimization, mutation, crossover, and selection strategies.
- Integrated **Computer Vision** and **OCR** (OpenCV, Tesseract) to automatically extract Sudoku puzzles from images and solve them in real time.
- Implemented algorithm benchmarking with **Backtracking vs Genetic Algorithm** comparison, convergence visualization, difficulty prediction, and performance analytics dashboard.

DeepWire DPI – Smart Network Traffic Analyzer *In Progress C++, libpcap, TCP/IP, Computer Networks*

- Built a Deep Packet Inspection (DPI) engine using **libpcap** for real-time packet capture, parsing, and protocol analysis.
- Developed traffic classification and **TLS SNI extraction** modules for network monitoring and domain-level traffic intelligence.

Spanzia – Budget Tracking Mobile Application *React Native, TypeScript, Firebase*

- Developed a cross-platform personal finance application with expense tracking, budget management, and real-time transaction monitoring.
- Integrated **Firebase Authentication** and **Firestore** for secure user management, cloud synchronization, and scalable backend services.

Bloom Branding – Full Stack Web Platform *Next.js, TypeScript, Firebase*

- Engineered a full-stack web platform during **Google Winter of Code (GWOC)** using Next.js and Firebase.
- Implemented authentication, dynamic content management, and real-time data handling to improve user engagement and scalability.

Technical Skills

Languages: Python, C++, JavaScript, SQL

Frameworks: React, Next.js, Flask, React Native

Core Concepts: Data Structures, Algorithms, OOP, DBMS, Computer Networks

Tools: Git, GitHub, Firebase, VS Code

Achievements

- **Genesis Bootcamp:** Top 3 — built a full-stack expense tracking application.
- **ACM Summer Challenge:** Finalist after 5 rounds of DSA and optimization-based contests.
- **MindSprint Hackathon:** 48-hour hackathon — built web platform with backend services.

Positions of Responsibility

Nexus (CSE Departmental Cell), SVNIT *Representative* Aug 2024 – Present

- Organized technical coding events, mentorship sessions, and orientations for 200+ students.
- Collaborated with teams to manage event execution and technical content.

Mindbend (Technical Festival, SVNIT) *Co-Head, Web Development* Jan 2025 – Present

- Leading frontend development and architecture of the official festival website using modern frameworks.
- Executive, Infrastructure* Jan 2025 – Sept 2025
- Coordinated logistics and infrastructure setup for large-scale technical events.